



LEARNING MANAGEMENT SYSTEM
FOR **SJB** and **ICAS**

PROPOSAL

CREATE. DESIGN. PUBLISH.

PREPARED BY:

Mark Francis De Guzman

Claire Abiera

Maribeth Abrigo

Joshua Domingo

Andrei Yuan Gonzaga

Mariket Custodio

PREPARED FOR:

Ruel Savandal

IT Manager

ILITS



EXECUTIVE SUMMARY

This project aims to design and develop a web-based Learning Management System (LMS) tailored to the specific academic and administrative needs of SJB and ICAS. The system will streamline and digitize key educational processes, resulting in eliminating some of the manual processes of enrollment, student registration, and more.

HISTORY

Both SJB and ICAS have long depended on traditional, paper-based systems to manage essential academic and administrative operations. While this has worked in the past, the increasing number of students and subjects over the years has made it difficult to maintain accuracy and organization.

Faculty and staff have expressed challenges in keeping up with manual encoding, locating student records, and coordinating subject loads. These inefficiencies have highlighted the need for a more modern, digital approach.

SOLUTION

To address these needs, we propose the development of a custom Learning Management System. The LMS will automate student enrollment, registration, subject assignment, and grade management—all accessible through a web browser. It will support multiple user roles, offer intuitive dashboards, and ensure data security through proper authentication and role-based permissions.

By shifting these processes to a centralized digital platform, the LMS will significantly reduce the time and effort needed to manage academic tasks. Most importantly, it will allow students, faculty, and administrators to focus more on learning and teaching, and less on paperwork and manual encoding.



PROJECT BACKGROUND

With the continuous growth of student populations at SJB and ICAS, the limitations of traditional, manual academic processes have become increasingly evident. Current systems for student registration, subject assignment, grade tracking, and faculty coordination are predominantly paper-based or managed through disconnected tools. These practices often result in delays, data inconsistencies, and administrative bottlenecks.

In today's digital age, schools are expected to modernize their systems to align with global educational trends. Web-based learning platforms are now seen as essential for improving efficiency, transparency, and accessibility. By adopting a Learning Management System (LMS), SJB and ICAS have the opportunity to improve their overall academic and administrative performance while also enhancing the user experience for students, faculty, and administrators.

PROJECT OBJECTIVES AND GOALS

GENERAL OBJECTIVE:

To design and develop a fully functional web-based Learning Management System that meets the academic and administrative needs of SJB and ICAS.

SPECIFIC GOALS:

- Automate student registration and enrollment workflows
- Digitally manage course offerings, subject lists, and faculty assignments
- Enable faculty to submit and update student grades securely
- Allow students to view their subjects, schedules, and grades in real-time
- Provide administrators with centralized control over user access, academic records, and reports
- Ensure system usability, accessibility, and scalability for future enhancements



SCOPE AND METHODOLOGY

SCOPE OF WORK:

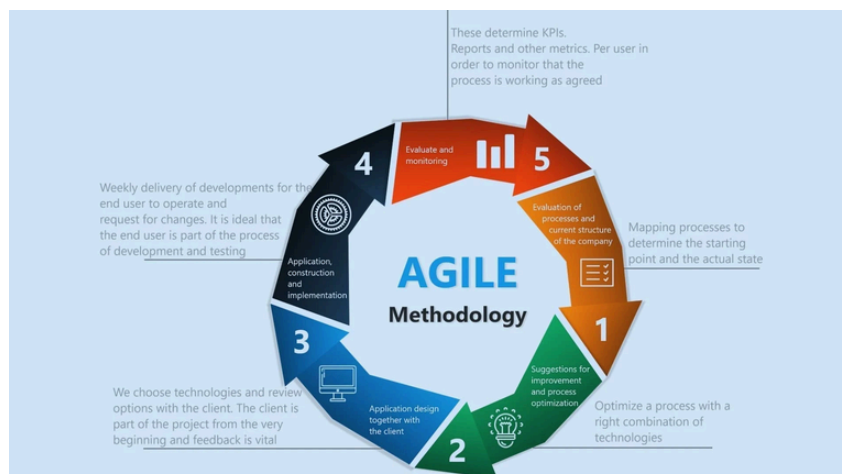
- Student registration and enrollment module
- Course and subject management interface
- Faculty registration and subject assignment module
- Grade submission and viewing portal
- Admin dashboard with user-role management
- Secure login system with session control
- Responsive front-end design compatible with desktop and mobile devices

TOOLS AND TECHNOLOGIES:

- Backend: PHP (MySQLi), MySQL
- Frontend: HTML, CSS, JavaScript, Bootstrap
- Authentication: Role-based access control with session management
- Deployment: Localhost/XAMPP for development, scalable to web hosting

METHODOLOGY:

The project will follow an **Agile** approach, with development divided into multiple sprints. Each sprint will deliver functional components that are tested and reviewed before integration.





TIMELINES AND MILESTONES

Phase	Duration	Target Date	Milestone
Project Planning & Research	1 week	Week 1	Requirements finalized
Database & Backend Setup	2 weeks	Weeks 2-3	Functional database and backend API ready
Frontend Design & Integration	2 weeks	Weeks 4-5	Responsive UI implemented
Core Module Development	3 weeks	Weeks 6-8	Enrollment, Subjects, Grades features built
Testing & Debugging	1 week	Week 9	Bug-free internal version
Final Revisions & Deployment	1 week	Week 10	Project completion and presentation



BUDGET AND RESOURCE ALLOCATION

Estimated Total Budget: ₱30,000 (adjustable based on school requirements)

Category	Estimated Cost
Web Hosting & Domain	₱5,000
Developer Compensation	₱15,000
Testing & QA Tools	₱2,000
Design & UI Resources	₱3,000
Contingency	₱5,000

Resource Allocation:

- Personnel: 1 project developer, 1 designer (if needed), 1 QA/tester
- Technology: Use of open-source technologies to minimize cost (XAMPP, Bootstrap, GitHub)

Time: Estimated 10 weeks of development



RISK ASSESSMENT AND MITIGATION STRATEGIES

Risk	Impact	Mitigation Strategy
Delay in gathering requirements	Medium	Early consultation with school stakeholders
Technical bugs or system errors	High	Regular testing after each development sprint
Scope creep (additional features)	Medium	Lock core scope early; evaluate additions later
Limited budget	Medium	Use cost-effective open-source tools
User resistance to change	Low-Medium	Offer system orientation and user manuals



EVALUATION AND SUCCESS METRICS

To ensure the success of the LMS project, we will measure outcomes using the following **Key Performance Indicators (KPIs)**:

Metric	Target
System Completion	100% of planned modules functional
User Adoption Rate	90%+ students and faculty onboarded
On-Time Delivery	All milestones met within timeline
Bug Rate Post-Launch	< 5 minor bugs reported
Admin & Faculty Satisfaction	Positive feedback from at least 80%
System Uptime	99% (post-deployment phase)



ORGANIZATIONAL CHART

